

STA 131A – Homework 4

Submission due: Tue, May 5 at 11:59 PM PT

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Instructions: Upload a single PDF file to Gradescope via Canvas (“Homework 4” under “Assignments”). Name the file using the prefix of your UC Davis email ID and the homework number (e.g., `dgsong_hw4.pdf`). Include “STA 131A,” your name, and the last four digits of your student ID on the front page. No late submissions will be accepted; any submission received after the deadline will receive 0 points. For full information about submission requirements and the late submission policy, see the syllabus.

Problem 1 (25 points in total). Investment portfolio under market uncertainty

The market can be in one of three possible states during the coming year:

$$M_1 = \{\text{prosperity}\}, \quad M_2 = \{\text{normal year}\}, \quad M_3 = \{\text{recession}\}.$$

Suppose

$$P(M_1) = \frac{1}{4}, \quad P(M_2) = \frac{7}{12}, \quad P(M_3) = \frac{1}{6}.$$

Let R denote the annual portfolio return, measured in percentage points. Thus, $R = 10$ means a return of 10%. Let

$$B = \{R \geq 10\}$$

denote the event that the annual return is at least 10%.

Suppose also that

$$P(B \mid M_1) = 0.9, \quad P(B \mid M_2) = 0.7, \quad P(B \mid M_3) = 0.2.$$

(a) (5 points) Compute $P(B)$, the probability that the annual return is at least 10%.

(b) (5 points) Given the observation of B , compute the posterior probabilities

$$P(M_1 \mid B), \quad P(M_2 \mid B), \quad P(M_3 \mid B).$$

Which market state is most likely, given B ?

(c) (5 points) For each market state, suppose that the portfolio return R has the following conditional distribution:

$$P(R = r \mid M_1) = \begin{cases} 0.2, & r = 30, \\ 0.3, & r = 20, \\ 0.4, & r = 15, \\ 0.1, & r = 5, \\ 0, & \text{otherwise,} \end{cases}$$

$$P(R = r \mid M_2) = \begin{cases} 0.2, & r = 20, \\ 0.5, & r = 10, \\ 0.3, & r = 5, \\ 0, & \text{otherwise,} \end{cases}$$

$$P(R = r \mid M_3) = \begin{cases} 0.2, & r = 10, \\ 0.5, & r = -10, \\ 0.3, & r = -40, \\ 0, & \text{otherwise.} \end{cases}$$

Compute the conditional expectation $\mathbb{E}[R \mid M_i]$ for each $i \in \{1, 2, 3\}$.

(d) (5 points) Use the law of total expectation to compute the unconditional expected return $\mathbb{E}[R]$.

(e) (5 points) Compute the conditional expectation

$$\mathbb{E}[R \mid B].$$

Hint: For each market state M_i , first compute $\mathbb{E}[R \mid M_i, B]$, and then use the posterior probabilities from part (b).

Problem 2 (35 points in total + 5 bonus points). Sampling from an urn

An urn contains 3 white balls and 7 black balls. Unless otherwise stated, treat the 10 balls as individually distinguishable.

- (a) **(9 points)** Suppose first that three balls are drawn sequentially without replacement.
- (i) How many ordered sequences of 3 distinct balls are possible, assuming that the 10 balls are individually distinguishable?
 - (ii) How many ordered length-3 color sequences are possible?
 - (iii) How many ordered sequences of distinct balls from part (i) produce the color pattern (Black, Black, Black)? How many produce the color pattern (White, Black, White)?

- (b) **(9 points)** Now suppose that five balls are drawn sequentially without replacement.

- (i) How many unordered 5-subsets of distinct balls are possible?
 - (ii) How many ways are there to split the 10 balls into two *unlabeled* groups of 5 balls each?
 - (iii) How many different color compositions are possible for a 5-ball selection?
- Hint:* How many different values can the number of white balls among the 5 selected balls take?

- (c) **(8 points)** Now suppose that three balls are selected uniformly at random without replacement, meaning that every 3-ball subset of the 10 individual balls is equally likely to be chosen. Let

X = the number of white balls selected.

Find the PMF of X , and compute $\mathbb{E}[X]$.

Hint: For $k = 0, 1, 2, 3$, count the number of ways to choose k white balls and $3 - k$ black balls.

- (d) **(9 points)** Now suppose the labels $0, 1, \dots, 9$ are assigned to the balls, and that balls of the same color are regarded as *indistinguishable* once the labels are assigned. Equivalently, a labeling is completely determined by choosing which 3 of the 10 labels are assigned to the white balls.

- (i) How many distinct labelings are possible?
- (ii) If all distinct labelings are equally likely, what is the probability that the white balls receive the labels $\{3, 6, 9\}$?
- (iii) What is the probability that all three white balls receive odd labels?

- (e*) **(5 bonus points)** Now consider a new urn containing 3 red balls, 5 yellow balls, and 7 green balls. Unless otherwise stated, treat the 15 balls as individually distinguishable.

- (i) Suppose 5 balls are selected without replacement, and only the *color composition* matters. How many different color compositions are possible?

Hint: Equivalently, how many triples (r, y, g) of nonnegative integers satisfy

$$r + y + g = 5, \quad 0 \leq r \leq 3, \quad 0 \leq y \leq 5, \quad 0 \leq g \leq 7?$$

- (ii) Suppose that all 15 balls are split uniformly at random into three *unlabeled* groups of 5 balls each.
 - (a) How many such groupings are possible?
 - (b) In how many of these groupings do the three red balls lie in three different groups?

Problem 3 (20 points in total + 5 bonus points). Joint and conditional PMFs

The joint PMF of two discrete random variables X and Y is

$p_{X,Y}(x,y)$	$y = 0$	$y = 1$	$y = 2$
$x = 0$	c	c	0
$x = 1$	c	$2c$	c
$x = 2$	0	c	c

and $p_{X,Y}(x,y) = 0$ for all other pairs.

- (a) **(6 points)** Find the value of c . Then compute the marginal PMFs p_X and p_Y .
- (b) **(6 points)** For each $x \in \{0, 1, 2\}$, compute the conditional PMF $p_{Y|X}(y | x)$ and the conditional expectation $\mathbb{E}[Y | X = x]$.
- (c) **(8 points)** Use the law of total expectation to compute $\mathbb{E}[Y]$. Then compute $\text{Var}(Y)$. Finally, determine whether X and Y are independent, and justify your answer.
- (d*) **(5 bonus points)** For each $y \in \{0, 1, 2\}$, compute

$$\mathbb{E}[X | Y = y] \quad \text{and} \quad \text{Var}(X | Y = y).$$

Then use conditioning on Y to compute $\mathbb{E}[XY]$. Based on your answer, determine whether the identity

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

holds in this example.

Problem 4 (25 points in total).

- (a) **(5 points)** [BT08, Chapter 3, Problem 2, p. 184]
- (b) **(5 points)** [BT08, Chapter 3, Problem 6, p. 186]
- (c) **(10 points)** [BT08, Chapter 3, Problem 7, p. 186]

References

- [BT08] Dimitri Bertsekas and John N Tsitsiklis. *Introduction to probability*, volume 1. Athena Scientific, 2nd edition, 2008.